



**Graph-Based Trust Propagation for Service Discovery in Informal Labour Markets: A
Case Study of Kigali, Rwanda**

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[Link to template](#)

Table of Contents

List of Tables

List of Figures

List of Acronyms/Abbreviations

CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

1.2 Statement of the Problem

1.3 Aim and Objectives of the Study

1.4 Research questions

1.5 Project scope

1.6 Significance and Justification

1.7 Research Budget

1.8 Research Timeline

1.9 Definition of Terms

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

2.2 Conceptual Review

2.3 Historical Background of the Research Topic

2.4 Theoretical Review

2.5 Overview of Existing System

2.6 Review of Related Works (Empirical Review)

2.7 General Comments

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

3.2 Research Design

3.3 System Analysis

3.4 System Design

3.5 Development Tools

REFERENCES

List of Tables

Table 1: List of Acronyms/Abbreviations

Table 2: Research budget

Table 3: Functional requirements

Table 4: Non-functional requirements

Table 5: Summary of reviewed literature

Table 6: Development tools

List of Figures

Figure 1: Research timeline

Figure 2: Proposed model diagram — trust propagation through the graph

Figure 3: System architecture diagram

Figure 4: Entity relationship diagram

Figure 5: Job lifecycle flowchart

Figure 6: UML class diagram

Figure 7: Use case diagram

Figure 8: Sequence diagram of the escrow payment and rating flow

List of Acronyms/Abbreviations

Acronym	Meaning
API	Application Programming Interface
BFS	Breadth First Search
BNR	National Bank of Rwanda (Banque Nationale du Rwanda)
GDP	Gross Domestic Product
GNN	Graph Neural Network
ILO	International Labour Organization
MoMo	Mobile Money
MVP	Minimum Viable Product
NISR	National Institute of Statistics of Rwanda

PWA	Progressive Web Application
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SMART	Specific, Measurable, Achievable, Relevant, Time-bound
UML	Unified Modeling Language

CHAPTER ONE: INTRODUCTION

1.1 Background to the Study

Informal employment is not a marginal feature of African labour markets; it is the dominant mode of work. The International Labour Organization (ILO, 2024) estimates that informal employment accounts for approximately 85% of all employment across the African continent, while analysis by the Institute for Security Studies (2025) places the informal sector's contribution at as much as 62% of GDP in Sub-Saharan Africa. Non-agricultural informal employment alone represents roughly two-thirds of total employment in the sub-region. Informality is unevenly distributed: young workers, women, and those without formal educational credentials are disproportionately concentrated in informal and vulnerable forms of work characterised by irregular income, absence of legal protection, and heavy dependence on personal social networks to obtain work at all.

Rwanda mirrors this regional pattern closely. According to the National Institute of Statistics of Rwanda (NISR, 2024), 82.5% of Rwanda's employed population — approximately 3.26 million people — work in the informal sector. This employment is concentrated in market-oriented agriculture, wholesale and retail trade, and construction: sectors defined by skilled and semi-skilled manual labour including masons, electricians, plumbers, carpenters, mechanics, cleaners and hairdressers. Kigali, as Rwanda's primary urban and economic centre, has experienced rapid urbanisation and a growing service economy, yet the mechanisms by which residents locate skilled workers, and by which those workers locate paying clients, remain informal, undocumented, and dependent on word-of-mouth referral circulating within narrow social circles. Traditional approaches to this coordination problem have relied entirely on socially embedded trust. A client seeking a mason asks a neighbour, a relative, or a previous contractor for a recommendation. This mechanism is not irrational: it transmits genuine information about reliability and competence, and it carries social accountability, since the referrer's own standing is implicated in the referral. Its weakness is not accuracy but reach. Word-of-mouth referral is bounded by the size of an individual's personal network, is non-transferable to strangers, cannot be searched or queried, and systematically excludes workers who are new to a city, young, or otherwise outside established referral circuits.

All over the continent, various software solutions have appeared in the shape of digital labour marketplaces, for example Lynk in Kenya and SweepSouth in South Africa. These platforms go beyond the boundaries of personal networks by making worker profiles available and acting as go-between for payments. Yet they replace the kind of trust mechanism that has been there before with one that is based on an average star rating derived from the views of

strangers. Wood and Lehdonvirta (2023) show that this kind of replacement turns the relational and situational trust that workers build up through repeated personal interactions into a single, decontextualised rating. This study suggests a third way: instead of getting rid of word-of-mouth or substituting it with anonymous aggregate ratings, it computes word-of-mouth itself as a trust graph, so that trust can spread along the chains of previous relationships in a way that is tailored to each individual client.

1.2 Statement of the Problem

Firstly, informal and blue-collar workers in Kigali cannot achieve consistent access to work because job discovery is heavily tied to personal social networks rather than mediated by any searchable system. With 82.5% of Rwanda's workforce employed informally (NISR, 2024), the overwhelming majority of the labour force operates without any durable, transferable record of skill or work history. A worker's reputation exists only in the memory of previous clients and cannot be conveyed to anyone outside that circle. Consequently, a competent mason three neighbourhoods away is functionally invisible to a client who has no mutual acquaintance, regardless of demonstrated competence. This constraint falls hardest on new entrants to the urban labour market — young workers and recent migrants to Kigali — who have not yet accumulated the years of referral relationships that established workers depend upon.

Secondly, the digital platforms which have grappled with this particular discovery issue in African markets have represented trust as an aggregate rating that does not reflect how trust is actually constituted in informal economies and their commercial trajectories demonstrate that this representation has not been sufficient. Lynk, launched in Kenya in 2015, had by 2019 processed approximately 23,000 jobs and disbursed over USD 2.5 million to more than 1,300 workers (Agence Française de Développement, 2020); it nonetheless wound down operations and was acquired by Eden Life in 2022, having shut down its website and halted new services (TechCabal, 2022). SweepSouth, founded in South Africa in 2014, continues to operate in its home market but withdrew from both Nigeria and Kenya in 2022 (Technext, 2022). Both platforms rank workers by a single averaged score visible identically to every user. Wood and Lehdonvirta (2023) find that such systems reduce relational trust to a decontextualised metric and can undermine rather than support worker recognition. Aggregate ratings further depend on rating participation that platforms frequently fail to obtain and offer no mechanism at all for workers with no rating history, producing a persistent cold-start barrier precisely for the new entrants who most need access.

Thirdly, payment for informal jobs in Rwanda is unfettered and unreliable. Since agreements are verbal and unpaid, workers have little recourse if clients withhold payments and clients have no assurance that they will receive the work promised to them prior to the completion of the job. Neither party can turn to a third party who will hold funds in trust until the work is completed to satisfaction. The lack of enforceability of payment adds to the credibility problem: neither party has an incentive to deal with anyone outside their existing network, since each knows that they cannot expect the other to deliver on the deal.

This work introduces a software solution designed to tackle these three sub-problems simultaneously, by implementing a personalised graph-based trust propagation algorithm which infers a client-specific trust score for workers that the client has not directly interacted with, based on weighted paths of prior transactions and referrals; a rating capture mechanism

that is structurally coupled to the release of payments, thereby mitigating the issue of sparse ratings; and a mobile-money-integrated escrow facility that holds funds agreed upon from the point of engagement until confirmed completion. These three mechanisms are intended to widen the reach of informal-sector-based trust beyond the client's personal network, without undermining its relational nature.

1.3 Aim and Objectives of the Study

The aim of this study is to develop a web-based service discovery platform that applies graph-based trust propagation to model and extend word-of-mouth reputation among informal and blue-collar workers in Kigali, Rwanda, coupled with a mobile-money escrow payment facility, in order to address the discovery, trust representation, and payment enforcement gaps identified in the problem statement, and thereby to reduce the dependence of informal hiring on preexisting personal networks.

1.3.1 List of the specific objectives

- To review the literature on trust propagation algorithms, digital reputation systems, and

informal labour markets in Sub-Saharan Africa, and to collect data on hiring, rating, and payment-delay practices through a structured survey of approximately 100 informal workers and clients with ties to the Kimironko, Kwa Nayinzira, Remera, and Gikondo, areas, distributed via

Google Forms, supplemented by 10 to 12 short semi-structured interviews for qualitative depth, in order to validate assumptions that inform the synthetic dataset design, within the first four weeks of the project.

- To design and develop a web-based platform implementing a personalised graph-based

trust propagation algorithm, a bidirectional rating capture flow coupled to payment release, and a sandboxed mobile-money escrow payment module, delivered as a functional minimum viable product by the end of week twelve.

- To evaluate the solution against measurable metrics that address the identified gaps, specifically: trust score coverage for cold-start users relative to a flat-average baseline; rating capture rate under payment-coupled prompting relative to an unprompted baseline; correlation between computed trust score and observed repeat-hire behaviour; and escrow transaction completion rate, evaluated over a simulated transaction dataset and a live usability pilot by the end of week sixteen

1.4 Research questions

4. To what extent does a personalised, graph-based trust score provide usable trust information for clients and workers with no direct transaction history, compared with a conventional flat-average rating system?

5. Does coupling the rating prompt to the payment-release action measurably increase the proportion of completed jobs that yield a usable trust edge, compared with an unprompted standalone review request?

6. Does the computed trust propagation score correspond to observed repeat-hire behaviour within the transaction dataset, and therefore constitute a meaningful predictor of hiring willingness?
7. To what extent does a mobile-money escrow mechanism improve the reliability and timeliness of payment settlement compared with the prevailing cash-based informal practice?

1.5 Project scope

Geographical scope. Testing a platform across an entire country within a single trimester is not feasible given time, infrastructure, and resource constraints. This project therefore scopes its pilot to the Kimironko, Kwa Nayinzira, Remera, and Gikondo, areas of Kigali. These areas combine a dense concentration of informal traders, artisans and service workers with a growing base of urban clients, making them both representative of the target population and logistically feasible for a single student researcher to access directly

Participant and data scope. Evaluation of the trust propagation algorithm is conducted primarily against a simulated transaction dataset modelled on realistic informal-sector parameters. This method stems from precedent in the trust propagation literature; Massa and Avesani (2007), Guha et al. (2004), Kamvar et al. (2003) all evaluate their algorithms on existing or constructed datasets rather than a newly-recruited live user population due to the algorithm-level properties of propagation coverage and path decay requiring transaction-level volumes that a small pilot within a single trimester cannot realistically generate. A separate live usability pilot of 10-20 users drawn from the two target areas establishes that the platform functions as software end-to-end and collects qualitative feedback on the ease of comprehension and usability.

Technical scope. The minimum viable product is delivered as a responsive web application, optionally installable as a Progressive Web Application to accommodate low-bandwidth conditions, comprising five modules: user registration and verification, job posting and quotation, trust-ranked worker discovery, rating capture, and escrow payment. Payment integration is implemented against a sandbox environment, as live payment processing requires payment service provider licensing from the National Bank of Rwanda that is outside the scope of a student project. Excluded from this phase are: multi-country deployment, native mobile applications, automated dispute arbitration, and integration with government identity or taxation systems.

1.6 Significance and Justification

For informal and blue-collar workers, successful implementation is expected to extend the reach of accumulated reputation beyond the boundary of the personal network, allowing competence demonstrated to one client to become visible to that client's wider network. This is of particular consequence for new entrants who currently face a structural barrier to accessing work.

For clients, the platform is expected to remove much of the practical burden currently involved in finding and engaging informal labour. Rather than asking around, waiting on referrals, or travelling to locate a worker in person, a client can post a job and receive quotations from trust-ranked workers directly. Trust scores derived from the client's own network, rather than an anonymous aggregate, make it faster to identify a worker likely to be

reliable without having to personally vet a stranger. The quotation-and-accept mechanism removes the need for in-person price haggling, since workers submit a fixed quote that the client can compare and accept digitally. Escrow further removes the specific stress of paying upfront for work that is not yet done, since funds are held and only released on confirmed completion, offering a protection that a verbal agreement cannot.

Academically, this study applies an established class of trust propagation algorithms, developed and validated principally in peer-to-peer and Western e-commerce contexts, to a domain in which it has not previously been technically applied: geographically embedded informal labour markets in Sub-Saharan Africa. The empirical results will indicate whether relational trust modelling offers measurable advantage over aggregate rating in this setting.

1.7 Research Budget

The project is designed to rely predominantly on free tiers and open-source tooling. Primary data collection is conducted through a self-administered Google Forms survey rather than in-person scheduled interviews, which substantially reduces field costs relative to an entirely in-person data collection design. Development tooling (Node.js, React, Vite, MongoDB Community Edition, Git, VS Code), the Paypack sandbox environment, and academic literature accessed through the university library incur no cost.

1.8 Research Timeline

Use the Gantt chart to present how different components of the project will be implemented with respect to time.

1.9 Definition of Terms(Where necessary)

- Trust graph: a directed, weighted graph in which nodes represent platform users and edges represent a trust relationship from one user to another, generated from completed job ratings and signup referrals.
- Trust propagation: the inference of a trust value between two users who have no direct relationship, computed by traversing weighted paths through the trust graph.
- Node: a single user (client or worker) represented as a point in the trust graph.
- Edge: a directed, weighted trust relationship between two nodes.
- Decay factor: a multiplier (default 0.5) applied to an edge's contribution for each additional hop it is removed from the enquiring user, reflecting the attenuation of trust as it travels further from direct experience.
- Referred flag: a passive endorsement a client may attach to a completed job rating, signalling a stronger vouch than the numeric score alone and increasing the resulting edge's weight and persistence.
- Cold start: the condition in which a system has insufficient interaction history to produce a meaningful trust score for a given user.
- Escrow: a payment arrangement in which funds are held by a neutral third party (here, the platform, via a mobile-money provider) until agreed conditions are met.
- Tranche: a portion of a total payment released separately from the remainder — in this platform, a deposit tranche (for materials) and a final tranche (on confirmed completion).

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This review sought software-related literature in three connected areas: computational trust and reputation algorithms, particularly graph-based trust propagation methods; digital labour platform systems and their reputation architectures; and empirical studies of informal and gig labour markets in Sub-Saharan Africa. Literature was identified through structured keyword searching of Google Scholar, the ACM Digital Library, IEEE Xplore, SpringerLink, arXiv and MDPI, supplemented by institutional and policy sources from the ILO, NISR, the Center for Global Development and CGAP for contextual and statistical grounding. Search terms combined algorithmic vocabulary ("trust propagation", "trust-aware recommender", "graph neural network trust", "reputation system") with domain vocabulary ("gig economy", "informal labour", "blue-collar platform", "Africa", "Rwanda"). Approximately forty candidate sources were screened by title and abstract, from which a working set was retained on the basis of relevance to the research questions, credibility of venue, and recency. Priority was given to publications from 2020 onward, with a small number of pre-2020 works retained deliberately as foundational sources that define the algorithms under consideration and are necessary to trace the intellectual lineage of the field. The review also incorporated a search of the African Leadership University thesis repository via Librarika to establish whether the proposed approach duplicates prior institutional work; no prior capstone applying trust propagation to informal labour discovery was identified.

2.2 Conceptual Review

The independent variable in this study is the trust representation mechanism employed by the discovery platform: specifically, whether worker trustworthiness is presented to a client as a personalised, graph-propagated score or as a conventional flat aggregate rating. The dependent variables are trust score coverage (the proportion of client-worker pairs for which the system can produce a meaningful trust value), rating capture rate (the proportion of completed transactions that yield a usable trust edge), and hiring willingness as evidenced by repeat-hire behaviour.

Three concepts require definition. Trust, in the computational sense adopted here, is a directed and subjective expectation held by one party regarding the reliability of another, rather than an objective property of the trusted party. This subjectivity is the conceptual basis for personalised rather than global trust scoring. Trust propagation is the inference of such expectations across relationship paths, resting on the assumption that trust is at least partially transitive: if A trusts B and B trusts C, this constitutes evidence, though weaker evidence, that A may trust C. Reputation, by contrast, is an aggregate public signal computed across all raters and is identical for every observer; the distinction between propagated trust and aggregate reputation is central to this study's contribution.

The conceptual framework connecting these variables holds that informal labour markets are coordinated by socially embedded trust, that digital platforms extend market reach but conventionally substitute aggregate reputation for embedded trust, and that a graph-based propagation mechanism may extend reach while preserving the relational structure of the original trust mechanism.

2.3 Historical Background of the Research Topic

2.3.1 Origins and Early Context of the research problem

The computational trust problem originated in distributed systems research in the late 1990s and early 2000s, when peer-to-peer file-sharing networks required mechanisms to identify malicious peers in the absence of any central authority. Kamvar, Schlosser and Garcia-Molina (2003) formulated the EigenTrust algorithm to address this problem, computing a single global trust value for each peer by iteratively aggregating local trust ratings weighted by the rater's own trust, a formulation mathematically equivalent to finding the principal eigenvector of the network's normalised trust matrix. The same period saw the emergence of a parallel line of enquiry concerned with trust in social and semantic web contexts, in which the object of interest was not a global reputation but the trust of a specific individual toward another.

2.3.2 Key Milestones

Guha, Kumar, Raghavan and Tomkins (2004) formalised the propagation of both trust and distrust across relationship networks using data from the Epinions review platform, demonstrating that trust composes across paths in a manner amenable to matrix formulation while distrust does not compose as reliably. Ziegler and Lausen (2005) extended this line of work with the Appleseed model, proposing spreading-activation-based local trust propagation as a computationally lighter alternative to global algorithms such as EigenTrust, and argued explicitly for trust metrics that are personalised to the trust source rather than global. Massa and Avesani (2007) subsequently established the practical significance of propagation for the cold-start problem, reporting that trust-based propagation increased the proportion of users for whom any recommendation could be generated from approximately 3% under conventional collaborative filtering to over 90%, while maintaining comparable accuracy. This finding remains the principal empirical justification for adopting propagation in systems with sparse interaction data.

The field subsequently absorbed developments in graph representation learning. Fan et al. (2019) introduced a graph neural network architecture jointly modelling user-item interactions and user-user social relationships, demonstrating that combining relational and transactional signals outperforms either alone. Dawn, Das and Bandyopadhyay (2023) extended this direction by incorporating rater reliability weighting into a graph neural network trust model. Most recently, Liu et al. (2026) proposed TrustGTN, a heterogeneous graph neural network that treats trust evaluation as an edge classification rather than node classification problem, arguing explicitly that trust is a property of a relationship rather than of an individual, and learning automatically which relationship paths are most informative rather than requiring manual specification.

2.3.3 Historical Gaps and transition to the current study

Two features of this lineage are notable for the present study. First, the empirical settings have been consistently narrow: peer-to-peer file-sharing networks, and Western consumer review and e-commerce datasets, principally Epinions, Ciao and FilmTrust. Second, the applications have been overwhelmingly concerned with digital goods, online content and

remote services rather than with physically situated labour. A systematic literature review of eighteen studies on the influence of digital platforms on gig workers (Alauddin et al., 2024) identifies a corresponding scarcity of research addressing Sub-Saharan Africa and manual trades specifically, despite the region's exceptionally high rate of informal employment. The transition to the current study follows directly: the algorithmic machinery is mature and well validated, but has not been applied to a setting in which trust is geographically embedded, socially dense, and economically consequential in the manner characteristic of African informal labour markets.

2.4 Theoretical Review

2.4.1 Foundational Theories

The study is based on three theoretical positions. First, the theory of social capital and embedded economic action, which argues that economic transactions do not occur between abstract rational agents but are embedded in ongoing social relations that provide information and enforce obligation. Word-of-mouth hiring in Kigali is thus not an informational shortfall but a working institution; its limitation is in reach rather than reliability. This is the motivation for modelling it rather than replacing it. Granovetter's (1973) operationalisation of tie strength as a measurable variable — combining time investment, emotional intensity, mutual confiding, and reciprocal services — supplies the basis for treating an ordinary job rating and an actively endorsed "referred" flag as structurally different edge weights within the same graph, since the two represent different intensities of tie rather than different types of relationship. This is distinct from, and should not be conflated with, Granovetter's separate and more widely cited finding regarding the strength of weak ties in diffusing novel information such as job opportunities; the present study draws only on the tie-strength-as-variable concept, not on the weak-ties bridging argument, since the referred flag is a claim about signal intensity rather than about network bridging. The second is the transitivity assumption in computational trust, formalised by Guha et al. (2004), which states that trust is partially and attenuating transitive along paths of relationships. This assumption is what licenses propagation in the first place, and its attenuating character is what requires a decay factor. Golbeck (2005) shows empirically that inferred trust values in social networks degrade predictably with path length, which informs the decision to bound propagation depth in the proposed algorithm. The third is the distinction between relational and aggregate reputation articulated in the platform labour literature. Wood and Lehdonvirta (2023) argue that platform rating systems compress relational trust into decontextualised metrics, with consequences for worker recognition and bargaining position. This supplies the normative and sociological rationale for preferring a personalised representation. The theoretical framework adopted for this study therefore treats the trust graph as a computational representation of social capital, propagation as the operationalisation of attenuated transitivity, and personalisation as the preservation of relational rather than aggregate reputation

2.5 Overview of Existing System (Review of related technical solutions)

Two platforms constitute the closest technical solutions to the problem addressed here, and the commercial history of both is instructive. Lynk, launched in Kenya in 2015, connected vetted informal workers including electricians, tailors and chefs with households and

businesses through mobile and web interfaces. Workers underwent in-person vetting prior to listing; clients paid through the platform by card or M-PESA; and the platform guaranteed disbursement to the worker within twenty-four hours of job completion, retaining a service fee. By 2019 it had processed approximately 23,000 jobs and disbursed over USD 2.5 million to more than 1,300 workers, with participating workers reportedly experiencing substantial increases in monthly earnings (Agence Française de Développement, 2020). Despite these results, Lynk wound down operations and was acquired by the Nigerian firm Eden Life in April 2022; at the point of acquisition it had already shut down its website, halted new services, completed outstanding orders and assisted departing staff into other employment (TechCabal, 2022). Startup databases now record the company as inactive.

SweepSouth, founded in South Africa in 2014, applies a comparable model restricted to domestic and home-cleaning services, with a predominantly female workforce, and has been independently assessed by the Fairwork Foundation as comparatively fair in its treatment of workers. It remains in operation in South Africa, where it publishes an annual survey of domestic worker earnings drawing on several thousand platform workers, and reported a median monthly wage of R5,545 for platform workers in 2025 against R3,932 for domestic workers outside the platform. Its expansion beyond South Africa was nonetheless unsuccessful: the company withdrew from Nigeria in November 2022, citing unfavourable macroeconomic conditions, and also closed its Kenyan operation (Technext, 2022).

A third category of comparator consists of general-purpose classified and listing services used informally for the same purpose in Rwanda, including social media groups and classified websites. These provide discovery without any verification, reputation or payment mediation whatsoever, and are included in the comparison to establish the baseline against which any structured platform must improve.

2.5.1 Strength and Weaknesses of the Existing System(s)

Lynk's principal strength was its demonstration that a combined vetting and payment guarantee measurably improves informal worker earnings, establishing that the intervention class produces real economic benefit for workers. Its weaknesses, relative to the problem addressed here, were threefold: it operated only in Kenya and was built around M-PESA payment rails and Kenyan vetting norms; it employed centralised in-person vetting, which imposes an operational cost proportional to the number of workers onboarded and therefore does not scale; and its reputation representation was an aggregate rating identical for all viewers, which neither reflected nor extended the relational trust structure of the informal market.

Lynk's closure is not by itself evidence that aggregate rating caused its failure, and no such causal claim is made here: platform shutdowns are typically overdetermined by funding conditions, unit economics and operational cost structures, and the available public accounts do not isolate any single cause. It is nonetheless significant for this study that the most prominent African platform to address precisely this problem, having achieved demonstrable earnings gains for participating workers, did not prove sustainable, and that its trust and vetting architecture was operationally expensive by design. This suggests that the problem is not adequately solved by the existing approach, and that a trust mechanism which accumulates from ordinary platform activity rather than from centralised manual vetting merits investigation on cost grounds as well as on representational ones.

SweepSouth's strength lies in its attention to worker protections and in its demonstrated durability, having operated continuously in South Africa for over a decade. Its weaknesses for present purposes are its narrow service scope, since a single-category domestic cleaning platform does not generalise to the range of skilled trades that constitute the majority of Rwanda's informal workforce, and its limited transferability across markets, evidenced by its withdrawal from both Nigeria and Kenya. General classified services, finally, offer reach without trust, and in doing so illustrate precisely why discovery alone is insufficient.

Critically, none of the three operates in Rwanda, and none employs personalised trust propagation.

2.6 Review of Related Works (Empirical Review)

Beyond the algorithmic lineage traced in Section 2.3 and the direct platform comparators reviewed in Section 2.5, three further bodies of empirical work bear on this study's design. The first concerns reputation system design specifically within labour markets, rather than in e-commerce or content platforms generally. Kokkodis (2021) analyses 58,459 completed tasks on a major online labour platform and finds that conventional reputation systems, which uniformly aggregate scores across a worker's entire skill set, misrepresent workers whose competence varies by task type, and proposes a skillset-specific and time-decaying alternative. This finding directly supports the present study's departure from flat aggregate rating, and its use of a decay factor is a close structural parallel to the decay applied here to propagated trust with distance, though Kokkodis models decay over time rather than over network hops.

The second concerns the specific empirical conditions of African gig and informal labour markets. Dinika's doctoral research into digital platformisation across Rwanda, South Africa and Zimbabwe, comprising over sixty semi-structured interviews with platform workers, documents how algorithmic management systems can undermine worker autonomy while also finding that workers retain meaningful agency to contest unfavourable platform conditions. This is directly relevant to the design choice, made in this study, to give workers a rating channel toward clients rather than only receiving ratings passively: a bidirectional mechanism of the kind proposed here is one concrete way of extending worker agency within a platform architecture, rather than leaving workers as the sole object of evaluation. Alauddin et al.'s (2024) systematic review of eighteen studies on digital platforms and gig workers, drawn from the Scopus and Web of Science databases under PRISMA guidelines, corroborates the pattern identified in Section 2.3.3: platforms are consistently found to offer low barriers to entry and flexible engagement, but to expose workers to a lack of social protection and downward wage pressure where reputation and pricing mechanisms are opaque or centrally controlled.

The third concerns the specific mechanics of trust propagation at a technical level. Ziegler and Lausen's (2005) Appleseed model, introduced in Section 2.3.2, additionally offers a methodological precedent for bounding propagation depth and normalising trust mass across outgoing edges, both of which inform the implementation described in Chapter Three. Taken together, this body of empirical work supports three specific design decisions in this study: personalisation over aggregation (Kokkodis, 2021), the inclusion of a worker-facing rating

channel rather than a purely client-facing one (Dinika, n.d.), and a bounded, decaying propagation mechanism rather than an unbounded global computation (Ziegler & Lausen, 2005).

2.6.1 Summary of Reviewed Literature

2.7 General Comments

The literature reviewed establishes, first, that graph-based trust propagation is a mature and empirically validated algorithmic approach, but one developed and tested almost exclusively on peer-to-peer file-sharing networks and Western e-commerce or content platforms; second, that the two African digital labour platforms closest in purpose to the present study both represented trust as an aggregate rating, and that this representation is criticised in the platform-labour literature on both practical and normative grounds; and third, that empirical research specific to African informal and gig labour markets is comparatively scarce, and where it exists, tends to focus on platform governance and worker experience rather than on the technical design of the underlying reputation mechanism. The gap this study addresses sits precisely at that intersection: a technically rigorous, graph-based trust propagation approach, applied for the first time to a geographically embedded informal labour market in Sub-Saharan Africa, and specifically to Kigali, Rwanda.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter presents the design of the proposed platform. It describes the research design and software development lifecycle model adopted; the baseline and proposed trust models, including the graph-based trust propagation algorithm at the centre of this study; the functional and non-functional requirements of the system; and the full system design, comprising the architecture, database design, flowchart, class diagram, use case diagram and sequence diagram. Development tools are set out at the end of the chapter.

3.2 Research Design (including the SDLC model used)

This project follows an Agile software development lifecycle, structured in two-week sprints across the sixteen-week timeline set out in Section 1.8. Agile was selected over a waterfall approach because the trust propagation algorithm's parameters (decay factor, propagation depth, referred-flag weighting) are expected to require empirical tuning once tested against the synthetic dataset, which is more naturally accommodated by iterative development with regular working increments than by a single up-front design phase. Data collection and requirements validation (Objective 1) run concurrently with early sprints rather than strictly preceding them, so that survey and interview findings can still inform design decisions made in the first half of the project.

Data collection for this study combines a structured survey of approximately 100 respondents, administered through Google Forms and distributed via convenience sampling through school networks, community WhatsApp groups, and referrals within Kimironko, Kwa Nayinzira, Remera, and Gikondo, with 10 to 12 semi-structured interviews selected through purposive sampling to ensure both client and worker perspectives are represented. The survey is designed to establish base rates — how residents currently find workers, how often payment disputes occur, and how willing residents are to trust a platform-mediated recommendation — that inform the parameters of the synthetic dataset described in Section 3.2.1. The interviews are designed to surface the qualitative texture behind those numbers: specifically, what a referral actually communicates in practice, and what would make a client or worker trust a digital recommendation over an in-person one, given that a self-administered survey cannot fully capture this kind of tacit reasoning.

3.2.1 Baseline Model Description

The baseline against which the proposed model is evaluated is a conventional flat-average rating system, structurally equivalent to the reputation mechanism used by Lynk and SweepSouth as reviewed in Section 2.5: every worker's trust representation is a single average of all ratings received, visible identically to every client regardless of that client's own network or transaction history. This baseline is used to compute the comparative trust score coverage and repeat-hire correlation metrics defined in Section 3.2.3.

3.2.3 Proposed Model Description

The proposed model computes a trust value that is specific to the ordered pair of enquiring party and candidate counterparty, and the graph is symmetric with respect to direction of use: the same underlying structure is queried to rank workers for a client and to surface a client's reliability to a worker. The trust graph is a directed weighted graph in which nodes represent

users, of either role, and edges represent trust relationships. Two edge types are generated: rating edges, created when either party completes and rates the other following a job, with weight derived from the rating score normalised to the interval zero to one; and referral edges, created when a user joins the platform through the invitation of an existing user, assigned a fixed base weight. Rating edges are recorded in both directions independently: a client's rating of a worker and a worker's rating of the same client are separate edges, generated at different points in the job lifecycle, and neither is contingent on the other having been submitted.

To compute a trust score for a specific client-worker pair with no direct edge between them, the system performs a weighted breadth-first traversal outward from the client, to a maximum depth of three hops. At each hop, the contribution of a path is the product of the edge weights along it, multiplied by a decay factor (default 0.5) raised to the power of the number of hops, so that trust inferred through more intermediaries counts for progressively less. Where a client has explicitly flagged a prior rating as "referred" — a passive endorsement stronger than the numeric score alone — the corresponding edge's weight is boosted by a fixed multiplier (1.4×) and its decay rate is reduced (0.65 rather than 0.5), so that a strongly endorsed connection both counts for more and persists further through the graph. Where multiple paths connect the same client-worker pair, the strongest path determines the resulting score.

Because propagation cannot produce a value where no path exists, the model incorporates a layered fallback. Where no path connects client and worker within the depth bound, the system substitutes a community-level trust value computed over workers within the same area, reflecting the geographic embeddedness of informal reputation. Where neither a path nor community history exists, the platform verification status supplies a baseline floor value, ensuring that new workers are ranked below trusted workers but remain visible and reachable rather than excluded entirely. This layering directly addresses the cold-start condition identified by Massa and Avesani (2007), and is a design decision of consequence for the research problem, since excluding unrated workers would reproduce precisely the barrier to new entrants that the project seeks to remove. Because the same traversal is applied symmetrically, a worker deciding whether to accept a job from an unfamiliar client can be shown a comparable trust value computed over that client's history of ratings received from prior workers, subject to the same cold-start fallback, directly addressing the asymmetric payment risk identified in the problem statement.

3.2.4 Metrics for Evaluation

- Trust score coverage: the proportion of client-worker pairs in the synthetic dataset for which the proposed model returns a value derived from an actual path or community fallback, rather than the lowest verification floor, compared against the baseline model's fixed 100% coverage (since an average is always definable) but at uniform, non-personalised resolution.
- Rating capture rate: the proportion of completed jobs in the simulated dataset that yield a usable trust edge under payment-coupled prompting, compared with an unprompted baseline rate drawn from survey-reported willingness to leave a standalone review.
- Trust-behaviour correlation: the correlation between a computed trust score and simulated repeat-hire behaviour within the transaction dataset.
- Escrow completion rate: the proportion of simulated and pilot transactions that reach a

RELEASED state without dispute, and the distribution of time-to-release.

3.3 System Analysis

3.3.1 Functional Requirements of the System

ID	Requirement	Actor	Priority
FR-01	Users shall register with a name, phone number, role, and area	User	High
FR-02	Workers shall list one or more skill categories	Worker	High
FR-03	Users shall optionally provide a referrer at signup, creating a referral trust edge	User	Medium
FR-04	Clients shall post a job with a category, description, and area	Client	High
FR-05	Workers shall submit a quote against an open job	Worker	High
FR-06	Clients shall view workers ranked by personalised trust score for a given category and area	Client	High
FR-07	The system shall compute a trust score via graph traversal for any client-worker pair	System	High
FR-08	The system shall apply a layered fallback (community trust, then verification floor) where no path exists	System	High
FR-09	The system shall require escrow funding before a job transitions to active state	System	High
FR-10	The system shall release a deposit tranche to the worker where a materials deposit was agreed	System	Medium
FR-11	Workers shall mark a job as complete	Worker	High
FR-12	Clients shall confirm completion, which shall require a rating input and release held funds	Client	High
FR-12b	Workers shall be able to rate the client after marking a job complete, independently of the client's own confirmation and rating	Worker	High
FR-13	The system shall auto-release held funds after a defined interval where the client does not confirm	System	High
FR-14	Either party shall raise a dispute, which shall lock held funds pending administrative review	Both	Medium
FR-15	The system shall write a trust edge upon each completed and rated job	System	High

FR-16	Administrators shall review verification submissions and disputes	Admin	Medium
FR-17	Clients shall optionally flag a completed job as 'referred', recording heightened endorsement independent of the numeric rating	Client	Medium

3.3.2 Non-functional Requirements of the System

ID	Requirement	Rationale and Target
NFR-01	Performance	Trust queries shall return within 500ms at pilot graph scale; discovery pages within 2s on a 3G connection
NFR-02	Usability	Interface shall be operable by users with limited digital literacy; core flows shall require no more than four steps
NFR-03	Accessibility	Responsive design with PWA installability to accommodate low-bandwidth and intermittent connectivity conditions
NFR-04	Security	Passwords hashed; escrow funds never held directly by the application, only by the payment provider
NFR-05	Data privacy	Personal data limited to what is functionally necessary; no sale or third-party sharing of user data
NFR-06	Reliability	Escrow status transitions shall be idempotent and recoverable from a failed webhook delivery
NFR-07	Scalability	Trust graph traversal shall remain within the performance target up to the full synthetic dataset scale ($\approx 1,000$ users)
NFR-08	Maintainability	Codebase organised by domain module (auth, jobs, trust, payments) to support iterative Agile development
NFR-09	Localisation	Interface text and currency formatting appropriate to a Rwandan Francs (RWF) context

3.3.2 Design Rationale

Several interaction-design choices in this system are deliberate responses to the population it serves, rather than default conventions, and are stated explicitly here rather than left implicit in the requirements above.

Short, minimal-step flows. NFR-02 requires that core flows take no more than four steps. This is a direct response to the population described in Section 1.1: a significant proportion of

prospective users conduct business primarily through spoken word-of-mouth rather than app-mediated workflows, and cannot be assumed to have the same baseline digital literacy or patience for multi-screen onboarding as a typical urban app user. Every additional step in a flow such as posting a job or accepting a quote is a point at which a first-time user with limited smartphone fluency may abandon the task altogether; keeping flows short is therefore a usability requirement with a direct bearing on adoption, not a stylistic preference.

Category-tier trust display rather than raw scores. The discovery interface shows a worker's trust position as one of a small number of labelled tiers (for example, "trusted in your network" or "trusted in this area") rather than exposing the underlying numeric trust value computed by the algorithm in Section 3.2.3. This follows directly from the theoretical distinction, drawn in Section 2.4.1, between relational and aggregate reputation: a decimal score invites exactly the kind of decontextualised, comparison-across-strangers interpretation that Wood and Lehdonvirta (2023) identify as the weakness of conventional platform ratings. A labelled tier communicates the same underlying signal — how this worker is connected to you — without asking the user to interpret a number whose meaning depends on graph structure they cannot see.

Structured counter-offer rather than open-ended negotiation. Price bargaining is a normal and expected part of hiring informal labour, and the platform does not attempt to remove it. What it removes is the open-ended messaging thread: a client may accept a worker's quote, decline it, or send back a single counter-amount, and the worker may then accept that counter-offer or submit one revised quote. Every step of this exchange is still a discrete recorded number rather than free text, which is the property that actually matters for later dispute resolution — it is the structure of the values, not the absence of negotiation, that keeps the agreed price legally and practically unambiguous.

Two further directions were raised in supervisory discussion and are deliberately scoped as future work rather than folded into the current MVP, for reasons worth stating explicitly. The first is attribute-based preferential search — for example, filtering workers by gender or other personal characteristics. This is a genuinely useful discovery feature in some cases (a client may have a legitimate reason to prefer a worker of a given gender for in-home work, for instance), but exposing arbitrary personal attributes as search filters also risks enabling discriminatory exclusion of workers on bases unrelated to competence. This study defers the feature pending a more careful definition of which attributes are appropriate to expose as filters and which are not, rather than implementing it without that consideration. The second is AI-assisted or semantic search — for example, allowing a client to describe a job in free text and be matched to relevant skill categories automatically, rather than selecting from a fixed category list. This is identified as a valuable extension to the discovery mechanism, but its implementation approach is not yet finalised, and building it credibly within the timeframe scoped in Section 1.5 alongside the trust propagation algorithm, which is the study's primary contribution, was judged an unrealistic addition to the MVP. Both directions are noted here as identified next steps rather than omitted silently.

3.3.4 Proposed Model Diagram customized to your mission

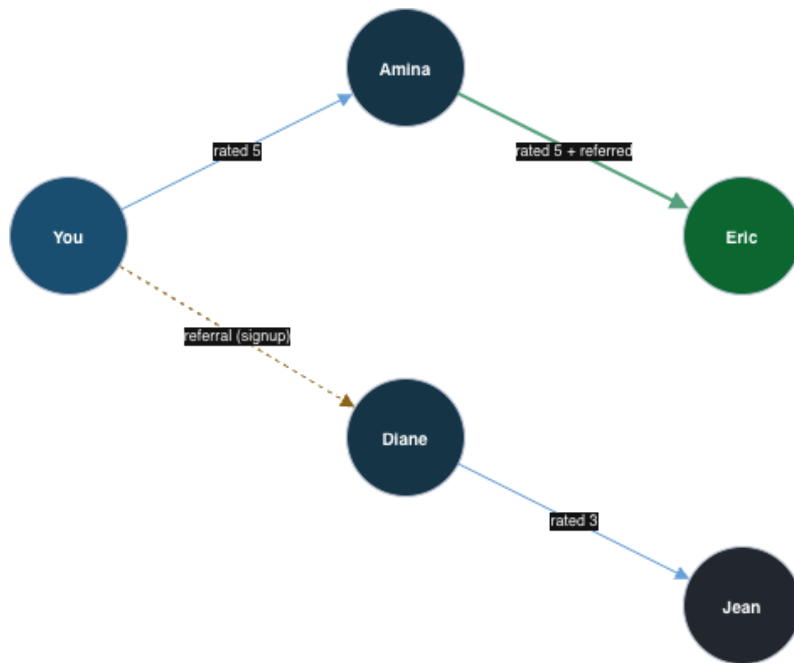


Figure 2: Proposed model diagram — trust propagation through the graph

3.4 System Design

3.4.1 Architecture Diagram

The system follows a conventional three-tier web architecture: a React and Vite single-page application on the client; an Express (Node.js) REST API; and a MongoDB database, with the trust propagation service implemented as an in-process module of the API rather than a separate service, given the pilot's scale. Escrow collection and disbursement are delegated entirely to Paypack, a mobile-money aggregator wrapping MTN Mobile Money and Airtel Money, so that the application itself never custodies funds directly — a design choice made specifically to avoid the payment service provider licensing requirement from the National Bank of Rwanda described in Section 1.5.

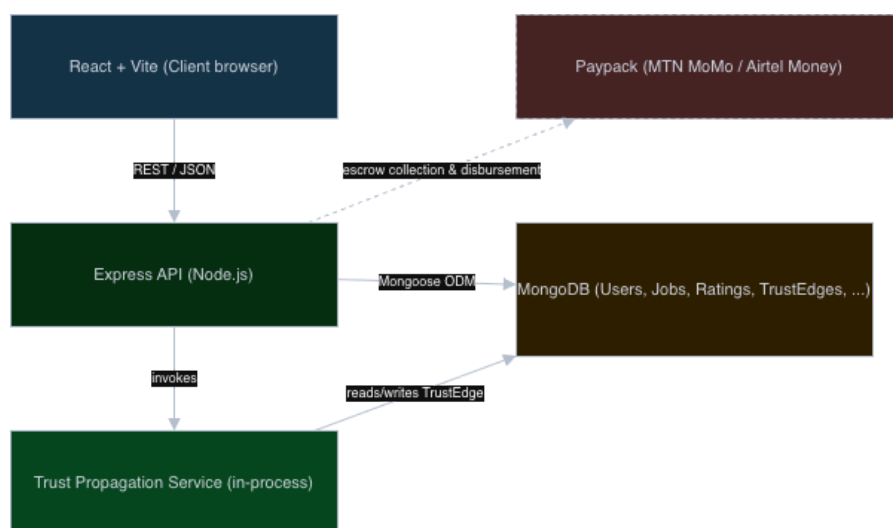


Figure 3: System architecture diagram

3.4.2 Database Design

The data model comprises six principal collections. The USER collection holds identity, role, sector and verification status. The JOB collection holds the posting, its status and the agreed price. The QUOTE collection holds worker quotations against a job, including any proposed materials deposit. The TRANSACTION collection holds escrow records, separated by tranche so that deposit and final payment are tracked independently while belonging to the same job. The RATING collection holds ratings in both directions: a client's rating of a worker and a worker's rating of a client are both stored as records distinguished by from_user and to_user, rather than as separate schemas, since a rating is structurally the same object regardless of which party submitted it. The TRUST_EDGE collection holds the directed weighted edges over which propagation is computed, generated from ratings and referrals, and indexed on source and target to permit efficient traversal in either direction.

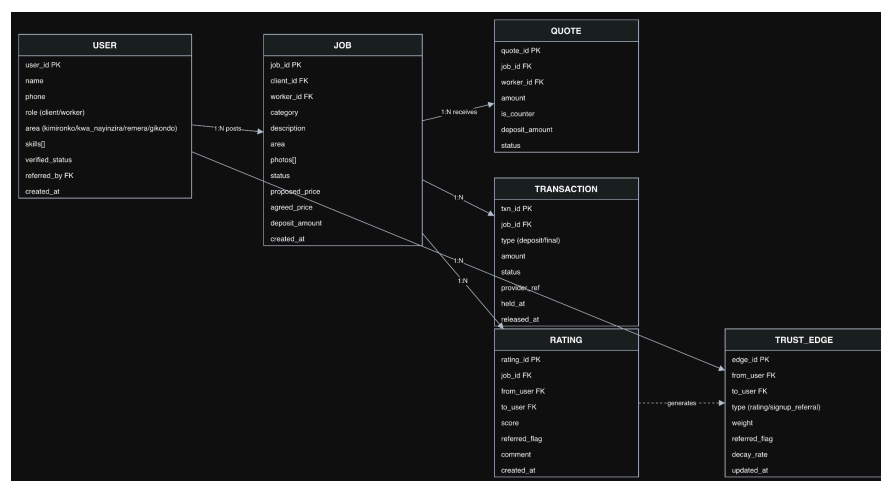


Figure 4: Entity relationship diagram

3.4.3 Flow Chart

Figure 5 traces a job from posting to completion, including the decision points governing materials deposits, disputes, and the timeout path that auto-releases funds where a client does not confirm. The worker-rates-client step is shown as a parallel, non-blocking branch, since it is independent of the client's own confirmation and rating.

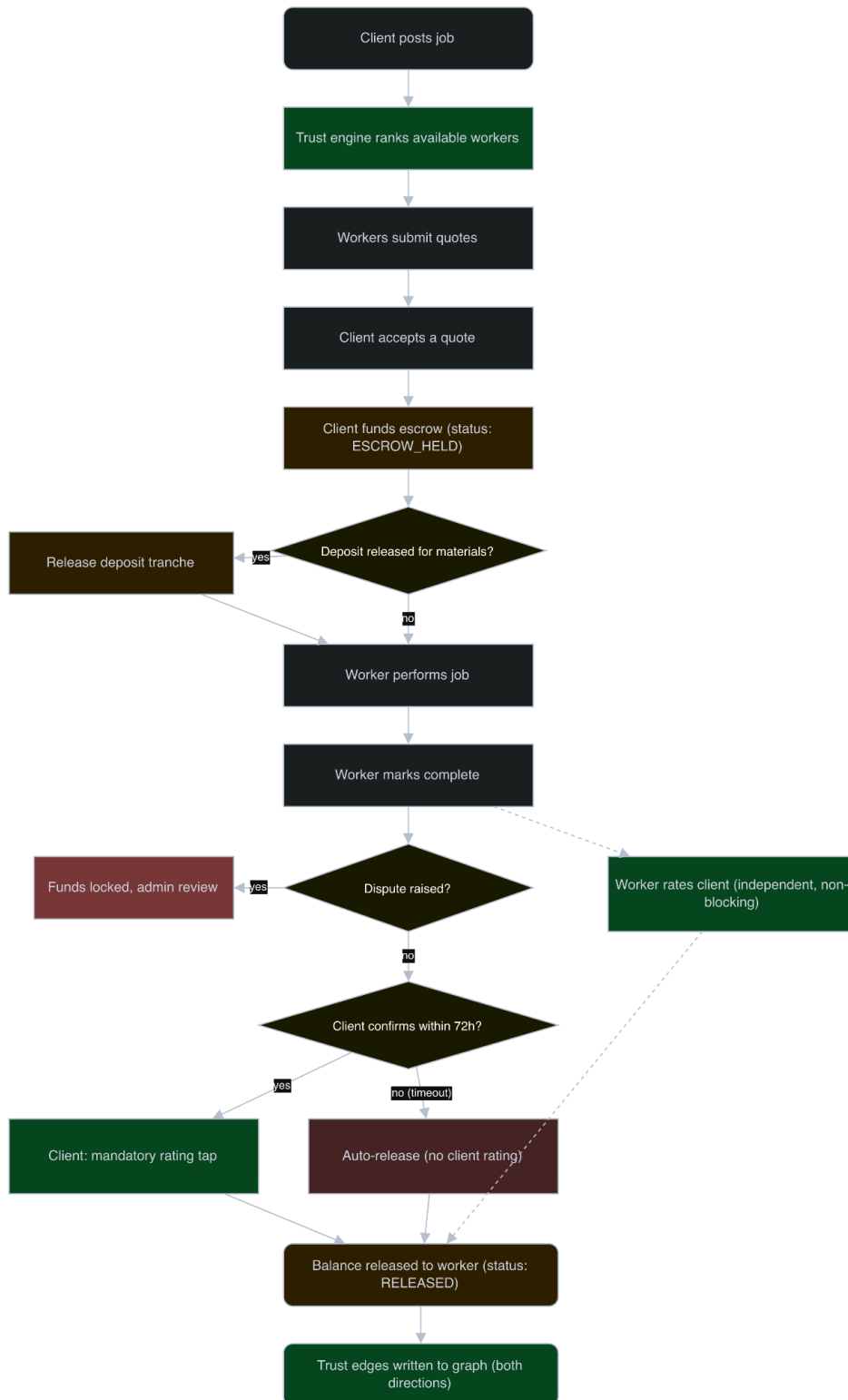


Figure 5: Job lifecycle flowchart

3.4.4 Class Diagram

Figure 6 presents the system's principal classes, corresponding directly to the Mongoose schemas implemented in the backend, together with the trust propagation service and its three primary operations.

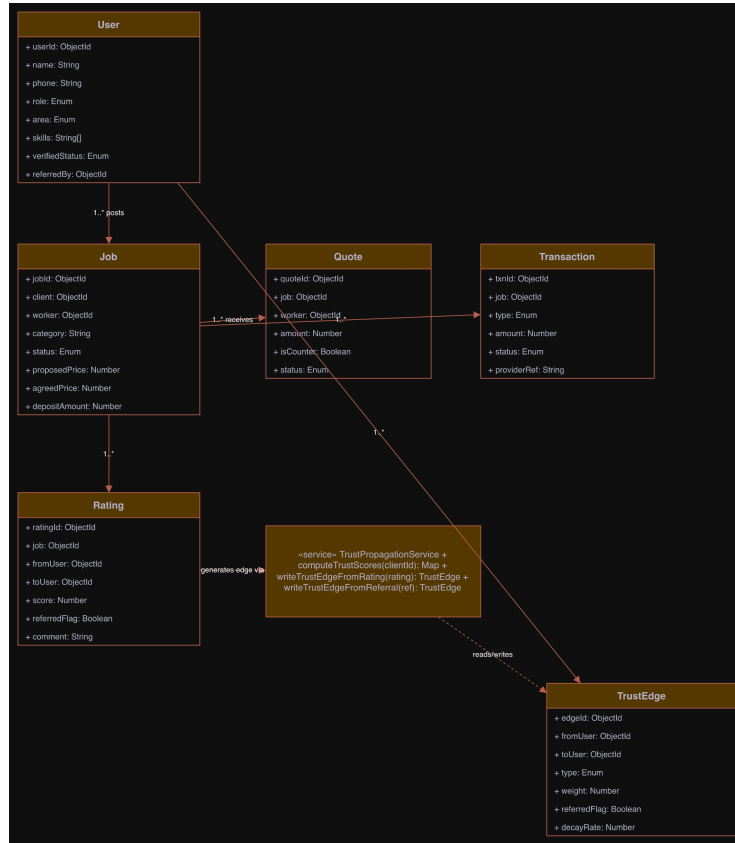


Figure 6: UML class diagram

3.4.5 Use Case Diagram

The system supports three actors. Clients post jobs, view trust-ranked workers, accept quotations, fund escrow, confirm completion and rate the worker. Workers register and seek verification, submit quotations, mark completion, rate the client independently, and receive payment. Rating is symmetric: both actors rate one another following a job, and neither rating is contingent on the other. The payment provider participates as an external actor in escrow funding and disbursement. Either party may raise a dispute.



Figure 7: Use case diagram

3.4.6 Sequence Diagram

The sequence diagram sets out the interaction between client, application server, database, payment provider and worker across the escrow and rating flow. Three aspects warrant note. First, the application server never holds funds: collection and disbursement are both performed by the payment provider, and the server controls only the timing of disbursement. Second, the client's rating is captured within the same request that triggers release, which is the structural basis of the rating capture intervention under evaluation. Third, the worker's rating of the client is submitted independently, upon marking the job complete, and its trust edge is written before the release sequence begins, so that a client's conduct in a given job is reflected in the graph regardless of how that same job later resolves.

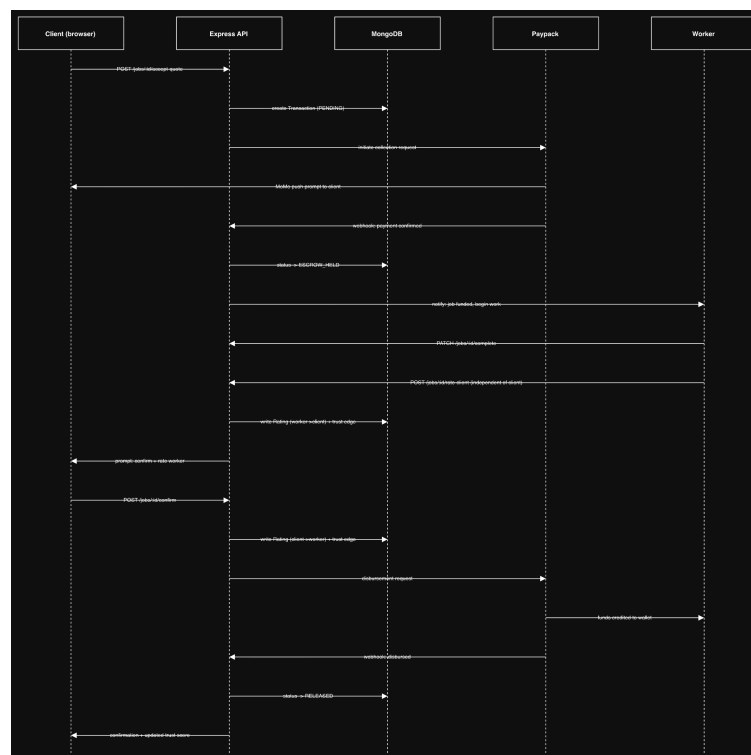


Figure 8: Sequence diagram of the escrow payment and rating flow

3.5 Development Tools

Category	Tool	Justification
Frontend	React with Vite	Component model suits the modular interface; Vite provides fast builds and a straightforward developer experience
Backend	Node.js with Express	Lightweight, widely supported, and consistent with the team's existing JavaScript proficiency
Database	MongoDB	Document model suits the platform's evolving, loosely-structured entities (jobs, quotes, ratings) better than a rigid relational schema at MVP stage
Payments	Paypack (sandbox)	Aggregates MTN MoMo and Airtel Money without requiring direct BNR payment-provider licensing
Version Control	Git and GitHub	Standard practice; enables incremental, reviewable
Design	Figma	Wireframing and UI design prior to implementation

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